CHAPTER FIVE

Genetics of Gallstone Disease

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Abstract

Gallstone disease (GSD) is one of the most common biliary tract disorders worldwide. The prevalence, however, varies from 5.9–21.9% in Western society to 3.1–10.7% in Asia. Most gallstones (75%) are silent. Approximately half of symptomatic gallstone carriers experience a second episode of biliary pain within 1 year. These individuals are at

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increased risk of developing acute cholecystitis, acute cholangitis, and biliary pancreatitis. As can be expected, these complications burden health care systems because of their invasive nature and surgical cost. Factors that contribute to gallstone formation include supersaturation of cholesterol in bile, gallbladder hypomotility, destabilization of bile by kinetic protein factors, and abnormal mucins. Epidemiologic studies have implicated multiple environmental factors and some common genetic elements in gallstone formation. Genetic factors that influence gallstone formation have been elaborated from linkage studies of twins, families, and ethnicities. Accumulating evidence suggests that genetic factors play a role in GSD.

ABBREVIATIONS

ABC ATP-binding cassette
ACAT acetyl-coenzyme A acetyltransferase
ADRB3 beta3-adrenergic receptor
ApoA-1 apolipoprotein A1
ATP adenosine triphosphate
BSEP bile salt export pump
CCK-1R cholecystokinin A receptor
CETP cholesteryl ester transfer protein
CYP7A1 cholesterol 7 alpha-hydroxylase
FXR farnesoid X receptor
GSD gallstone disease
GWA genome-wide association
HDL high-density lipoprotein
HMG-CoA 3-hydroxy-3-methylglutaryl-coenzyme A
LDL low-density lipoprotein
LXR liver X receptor
MDR1 multiple drug resistance protein 1
MUPCDH mucin-like protocadherin
NPC1L1 Niemann–Pick C1-like protein 1
QTL quantitative trait locus
RXR retinoid X receptor
SR-BI scavenger receptor class B type I
VLDL very-low-density lipoprotein

1. GENETIC BACKGROUND OF HUMAN CHOLESTEROL GALLSTONES

1.1. Epidemiology of cholesterol gallstones

Gallstone disease (GSD) is one of the most common biliary tract disorders worldwide. Epidemiologically, the highest age-adjusted prevalence (48%) has been found in Pima Indians, some North and South American Indians,

and Chileans [1–3]. Intermediate rates have been found in American (white), European (20%), and Asian populations (5–20%) [4,5], whereas Africans have the lowest incidence (<5%) [6]. A United States population-based survey showed that the prevalence of GSD was highest for Mexican Americans (27%) followed by non-Hispanic whites (17%) and non-Hispanic blacks (14%) among age-adjusted females. In contrast, age-standardized prevalence was similar for non-Hispanic white (8.6%) and Mexican American (8.9%) males. The prevalence for both was higher than that for non-Hispanic black men (5.3%). As such, genetics may play an important role in the predisposition to developing cholesterol gallstones [7]. Geographic and ethnic differences in gallstone prevalence imply that genetic factors influence risk of gallstone formation.

1.2. Family and twin studies

High familial prevalence of gallstones has been well recognized for at least four or five decades [8]. Cholesterol gallstones are more common by a ratio of 3:1 in siblings and other family members of affected persons than spouses or unrelated controls [9]. Sarin *et al.* studied 330 first-degree relatives of 105 gallstone patients [10]. This study found a gallstone incidence of 15.5%, which is approximately 4.5-fold higher than that for matched controls (3.6%) [10]. A similar result was also found in 202 first-degree relatives of subjects with gallstones (28.6%) versus 201 first-degree relatives of subjects without gallstones (12.4%) [11]. These findings clearly suggest that there is a strong familial predisposition for gallstone formation.

Twin studies have provided insight into the genetic epidemiology of complex traits and have been used to dissect complex genetics by analyzing the interaction of genotypes and phenotypes with age, gender, and lifestyle. Contrary to family studies, comparisons of the concordance rates of symptomatic gallstone disease (GSD) between monozygotic and dizygotic pairs of twins could determine whether the familial pattern is due to hereditary or environmental factors. Antero Kesaniemi *et al.* [12] investigated 35 Finnish twins and found that bile acid composition, cholesterol synthesis, bile cholesterol saturation, and gallstone formation were remarkably higher on pairwise correlations in monozygotic versus dizygotic males. This study highlighted the importance of genetic contributions to biliary lipid composition. Duggirala *et al.* [13] used pedigree data to explore the genetic susceptibility to symptomatic gallstones in 32 Mexican American families and estimated a heritability of $44 \pm 18\%$ for the proportion of phenotypic trait variance due to genetic effect. A large study of Swedish twins estimated

the contribution of genetic factors to the development of symptomatic GSD [14]. This report found that concordance rates in monozygotic twins of both genders (12%) were significantly higher than in dizygotic twins (6%) in 43,141 Swedish twins. Low concordance rates in monozygotic twins indicated the importance of environmental factors. They concluded that genetic effects accounted for 25% shared environmental effects for 13% and unique environmental effects for 62% of the phenotypic variance among twins. This data provided conclusive evidence for a strong genetic component in GSD.

1.3. Oligogenic predisposition to gallstones

Gallstone formation is a complex genetic trait that involves the effects of susceptible alleles of multiple *LITH* genes and their interactions with environmental factors. In specific groups of patients, mutations in single genes have been identified that result in a higher risk of gallstone formation. Pullinger *et al.* [15] found a link between a single-gene defect of cholesterol 7 alpha-hydroxylase (CYP7A1) and cholesterol gallstones associated with hypercholesterolemia resistant to 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors in two homozygote males. The homozygous deletion mutation in CYP7A1 gene could result in loss of enzyme function. CYP7A1 is a rate-limiting enzyme in the classical pathway for liver bile salt synthesis. As such, decreased bile salts in bile may lead to increased gallstone formation. Tangier disease, a monogenic disorder of familial high-density lipoprotein (HDL) deficiency, characterized by a biochemical defect in cellular cholesterol efflux, is caused by mutations in the ATP-binding cassette (ABC) 1 gene [16]. Missense mutations in the ABCB4 gene are another type of monogenic cholelithiasis [17] wherein cholesterol gallbladder stones and even recurrent cholesterol stones formed in bile ducts are associated with very low phospholipids/bile salt ratios in bile. Moreover, Asians with hepatolithiasis also exhibit low ABCB4 and hepatic phosphatidylcholine transfer protein causing markedly decreased biliary phospholipid secretion rates [18]. An aberrant splicing of cholecystokinin A receptor (CCK-1R), which is predicted to result in a nonfunctional receptor, was found in some obese patients with cholesterol gallstones [19]. CCK-1R-deficient mice show gallbladder hypomotility and prolonged small intestinal transit times, resulting in stasis of bile in the gallbladder and increased intestinal cholesterol absorption [20]. Wilson's disease is due to copper accumulation caused by mutations in the ATB7B gene [21] contributing to impaired liver function and increased hemolysis. A subset of patients with Wilson's disease have an increased risk of pigment stone formation. Progressive familial intrahepatic

cholestasis type 1, 2, and 3 are due to mutations in ATP8B1, ABCB11, and ABCB4, respectively. Mutations in ABCB4 can result in progressive cholestatic disease, while mutations in ATP8B1 and ABCB11 can result in episodic cholestasis that may induce bile stasis and subsequent gallstone formation [22].

The above genetic mutations were unraveled in a small number of genes affecting hepatic cholesterol metabolism, canalicular lipid transport, and motility of the gallbladder and the small intestine. Although these mutations do not explain susceptibility to gallstone formation in the majority of patients, they provide insight to the underlying pathophysiology of cholelithiasis.

1.4. Polygenic predisposition to gallstones and polymorphism studies

Human and mouse model studies have demonstrated that polygenes underlie the predisposition for most cholesterol gallstone development [14,23]. In the majority of cases, gallstones might develop as a result of lithogenic polymorphisms in several genes and their interactions with multiple environmental factors. Buch *et al.* performed a genome-wide association (GWA) study of >500,000 SNP in 280 individuals with gallstones and 360 controls. They identified a coding variant rs11887534 (D19H) of the sterol transporters ABCG5/G8 on the canalicular membrane of hepatocytes as a risk factor for cholesterol gallstone development [24]. The D19H variant of cholesterol transporter ABCG5/G8 was also identified as a genetic determinant of gallstone formation [25]. This study showed that the ABCG5/G8 gene confers an odds ratio (OR) of 2:3 in heterozygotes and 1:7 in homozygous carriers based on GWA [24] and linkage [25] studies.

Certain polymorphisms of the apolipoprotein (APO)-E genes, ApoB genes, and the cholesteryl ester transfer protein (CETP) are associated with gallstone development. ApoE is a major and minor protein component of VLDL (very-low-density lipoprotein) and HDL, respectively. Defects in ApoE result in familial type III hyperlipoproteinemia (HLP III) [26]. Impaired clearance of chylomicron, VLDL, and low-density lipoprotein (LDL) remnants results in increased plasma cholesterol and triglycerides. Certain polymorphisms of the ApoE genes correlate with cholesterol gallstone formation. ApoE is inherited as three allelic variants, ε2, ε3, and ε4. The ε2 genotype protects against gallstone formation in women [27]. The ε4 genotype is a genetic risk factor for cholelithiasis [28]. However, the association of ApoE ε4 alleles and gallstones was not confirmed in

subsequent studies [29]. Miettinen *et al.* demonstrated that intestinal cholesterol absorption was dependent on ApoE isoforms with ε4 > ε3 > ε2 [30]. Fecal excretion of cholesterol was higher in the ApoE ε2 phenotype versus the ε3 or ε4 phenotype.

Many genetic polymorphisms result in susceptibility to GSD. These include the cholesterol transporter [24,25,31–34], plasma transport [27,28,35–39], CETP [40] and uptake [41,42], bile acid synthesis [38,43,44], transporter [45–47] and bilirubin excretion [48,49], mucin [50,51], gallbladder motility [52,53], and hormone receptor [54] (Table 5.1).

1.5. QTL study and *LITH* genes

Conventional genetic mapping methods are designed for single-gene traits. As such, they are inadequate to study quantitative polygenic traits. Consequently, genetic analysis of affected sibling pairs or families is hampered by multiple modes of inheritance of the trait, genetic heterogeneity, and large variations in environmental conditions [56]. Linkage studies using quantitative trait locus (QTL) analysis have identified and localized genomic regions associated with gallstone susceptibility [57]. Genetic analyses of cholesterol gallstone formation using this inbred mouse model have provided insights to the complex pathophysiologic mechanisms that may facilitate identification of *LITH* genes in humans. However, mapping resolution is typically too coarse to identify the causal genes. Recent developments in high-density SNP genotyping and GWA SNP-based testing in outbred mice can also be applied to humans [58,59]. GWA analysis can be applied to outbred mice populations to identify genetic variants underlying complex traits. These studies can also complement classical linkage analyses using inbred mouse crosses to refine QTL and identify causal genes [60].

2. PATHOPHYSIOLOGY OF GALLSTONE FORMATION

Three pathologic abnormalities are considered relevant to gallstone formation [61]. These include cholesterol supersaturation in bile, nucleation, and precipitation of excess cholesterol from biliary micelles and gallbladder hypomotility. The primary pathophysiologic defect in cholesterol GSD is hypersecretion of hepatic cholesterol into bile. Hepatic cholesterol secretion into bile is affected by a lithogenic diet [62,63], intestinal absorption of cholesterol [64,65], a complex genetic predisposition, and *LITH* genes [66,67]. Bile becomes supersaturated with cholesterol when biliary concentration is increased or when the concentration of bile salts or

Table 5.1 Single nucleotide polymorphisms in gallstone disease

	Gene	Gene describe	SNP	Results	References
Cholesterol transporter	ABCG8	ATP-binding cassette, subfamily G, member 8. Transporter that appears to play an indispensable role in the selective transport of the dietary cholesterol in and out of the enterocytes and in the selective sterol excretion by the liver into bile	rs11887534 (D19H)	– $P=0.017$, odds ratio (OR) = 2.274 in Indian population – OR for D19H carriership is 2.2 in German – The risk of gallstones in carriers of the 19H allele was significantly increased in randomly selected cases from the ASP cohort compared to the stone-free controls (OR = 3.018; $P=0.017$)	[24,25,31–33]
			T400K	Male carriers of the less frequent K allele of ABCG8 T400K had a 2.31-fold elevated risk ($P=0.023$) for gallstone disease compared to males with the common genotype after the adjustment for age, body mass index.	[55]
	ABCG5	ATP-binding cassette, subfamily G, member 5	604Q	Increased risk of gallstone disease, adjusted OR = 4.7	[33]
Plasmatic transport	ApoA1	Apolipoprotein A-1. Major protein of plasma HDL, also found in chylomicrons. Synthesized in the liver and small intestine	−75G > A	Patients with the GG genotype ($P=0.015$) and G allele carriers ($P=0.004$) had a significantly higher risk of gallstone disease (1.087-fold and 1.561-fold, respectively)	[36]

Continued

Table 5.1 Single nucleotide polymorphisms in gallstone disease—cont'd

Gene	Gene describe	SNP	Results	References
ApoB	Apolipoprotein B (including Ag(x) antigen). ApoB-100 functions as a recognition signal for the cellular binding and internalization of LDL particles by the ApoB/E receptor	*Xba*I c.2488C > T, c.4154G > A	– The frequency of the X+/− genotype (20.63% vs. 7.94%) and X+ allele (10.79% vs. 3.97%) was significantly higher in the patient group than in the control group – Percentages of X+ allele 8.57% in GSD and 4.01% in healthy controls ($P < 0.01$)	[38,39]
		4154G > A (*Eco*RI)	A risk of gallstone formation was reduced in 4154AA homozygotes (OR = 0.25, $P = 0.009$) and heterozygous individuals (OR = 0.63, $P = 0.03$) as compared to 4154GG homozygotes	[35]
ApoC1	Apolipoprotein C-1. Appears to modulate the interaction of ApoE with beta-migrating VLDL and inhibit binding of beta-VLDL to the LDL receptor-related protein	*Hpa*I	Frequency of H2H2 was significantly higher ($P = 0.017$) in patients than in controls, and it was imposing a very high risk (OR = 9.416, 95% (confidence interval) CI = 1.125–78.786) for gallstone disease	[37]

	ApoE	Apolipoprotein E. Mediates the binding, internalization, and catabolism of lipoprotein particles. It can serve as a ligand for the LDL (ApoB/E) receptor and for the specific ApoE receptor (chylomicron remnant) of hepatic tissues	E2/3/4	– The E4/3 phenotype was enriched in both patients with gallstones and those who underwent cholecystectomy, with significantly ($P<0.006$) higher epsilon 4 allele frequencies than in gallstone-free subjects (OR = 2.67 and 3.62, respectively) – In women with apolipoprotein E2 (phenotypes E2/2, 2/3, and 2/4) compared with women without E2 (E3/3, 4/3, and 4/4), the OR for GSD was 0.28	[27,28]
Cholesteryl ester transfer protein	CETP	Cholesteryl ester transfer protein. Involved in the transfer of insoluble cholesteryl esters in the reverse transport of cholesterol	rs693, rs708272	– The nonancestral T/T genotype of apolipoprotein B rs693 polymorphism was associated with a decreased risk of GBC (OR: 0.14, 95% CI = 0.03–0.63). The T/T genotype of cholesteryl ester transfer protein (CETP) rs708272 polymorphism was associated with an increased risk of GBC (OR: 5.04, 95% CI = 1.43–17.8)	[40]

Continued

Table 5.1 Single nucleotide polymorphisms in gallstone disease—cont'd

	Gene	Gene describe	SNP	Results	References
Cholesterol uptake	LRPAP1	Low-density lipoprotein receptor-related protein associated protein 1, which plays a key role in cholesterol metabolism	Intron 5 insertion/deletion	– Frequency of Ins allele was significantly higher in the patient group than in the control group ($P=0.003$). Frequencies of Del and Ins allele were 65.77% and 34.23% in patients, 76.24% and 23.76% in controls – The D allele of LRPAP1 was significantly higher in GBC patients as compared to gallstone patients ($P=0.013$; OR = 1.6, 95% CI = 1.1–2.4)	[41,42]
Bile salt synthesis	CYP7A1	Cytochrome P450, family 7, subfamily A, polypeptide 1. Involved in lipid metabolism, and bile acid biosynthesis pathways	−204A > C	Percentages of A allele in patients and controls were 62.86% and 54.38% ($P<0.05$)	[38,44]
	FGFR4	Fibroblast growth factor receptor 4 (FGFR4) plays an important role in maintaining bile acid homeostasis by regulating the expression of cholesterol 7 alpha-hydroxylase (CYP7A1), a rate-limiting enzyme for bile acid biosynthesis	Gly388Arg (G-388R)	The ratio of gallstone patients with acute cholecystitis in the FGFR4 RR genotype (42%) was significantly higher than that in other genotypes of FGFR4 ($P=0.019$)	[43]

Bile acid transporter	SLC10A2	Solute carrier family 10 (sodium/bile acid cotransporter family), member 2. Plays a critical role in the sodium-dependent reabsorption of bile acids from the lumen of the small intestine. Plays a key role in cholesterol metabolism	rs9514089	Male nonobese rs9514089 was highly significantly linked to cholelithiasis ($P=0.00767$, OR $=2.04$)	[46]
			rs9514089	Association with low plasma cholesterol levels ($P=0.05$).	
	SLCO1B1	Solute carrier organic anion transporter family, member 1B1. Mediates the Na(+)-independent transport of organic anions Organic anion transport protein 1B1 is a major transporter protein for bile salt uptake in enterohepatic circulation of bile salts	rs11045819 (Exon4 C > A, Pro155 Thr); rs4149056 (Ex6 + 40T > C, Val174Ala)	The frequency of CA genotype and A allele of Exon4 C > A polymorphism was higher in gallstone patients (12.4% and 6.2%) as compared to controls (5.2% and 2.6%), which was statistically significant [($P=0.029$; OR $=2.31$; 95% CI $=1.1$–5.0); ($P=0.034$; OR $=2.22$; 95% CI $=1.1$-4.8), respectively]	[45]
	NR1H4 (FXR)	Nuclear receptor subfamily 1, group H, member 4. Receptor for bile acids such as chenodeoxycholic acid,	NR1H4_1 [T-G-A]	OR $=2.09$ (1.13–3.86); $P=0.02$	[47]

Continued

Table 5.1 Single nucleotide polymorphisms in gallstone disease—cont'd

	Gene	Gene describe	SNP	Results	References
		lithocholic acid, and deoxycholic acid. Represses the transcription of the cholesterol 7 alpha-hydroxylase gene (CYP7A1) and activates the intestinal bile acid-binding protein (IBABP). Activates the transcription of bile salt export pump ABCB11 by directly recruiting histone methyltransferase CARM1 within its gene locus			
Bilirubin excretion	UGT1A1	UDP glucuronosyltransferase 1 family, polypeptide A1 gene affects steady-state bilirubin levels and the incidence of gallstones in children with SCA Transforms bilirubin into water-soluble, excretable metabolites	Promoter	– UGT1A promoter polymorphisms may influence the ability of hydroxyurea to prevent gallstone formation in patients with SCA – Children with SCA had a lower frequency of the normal (TA)6 UGT1A promoter allele (0.413) than the abnormal (TA)7 allele (0.461)	[48,49]

Mucin	MUC1	Membrane-bound mucin	rs4072037	SNP rs4072037 at MUC1 was significant ($P=0.035$) in males	[50]
	MUC2	Gel-forming mucins	rs7396030	For males, the additive interaction model based on rs4072037 at MUC1 and rs7396030 at MUC2 yielded an age- and BMI-adjusted OR of 4.68 ($P=0.0008$)	[50]
	MUPCDH	Membrane-bound mucin	rs3758650	GSD association with an odds ratio (OR) of 1.59 (adjusted $P=0.013$) for the AG genotype and 5.82 (adjusted $P=0.007$) for the AA genotype	[51]
Gallbladder motility	CCK1R	Cholecystokinin receptor A mediates signals resulting in gallbladder contraction	CCK 27–33 sulf.	The frequency of the A1A1 genotype of CCK-AR was significantly higher in gallstone patients than in healthy individuals ($P=0.008$ odds ratio OR = 2.25, and 95% CI = 1.2–4.1)	[52]
	ADRB3	Adrenergic, beta-3-, receptor. Beta-adrenergic receptors mediate the catecholamine-induced activation of adenylate cyclase through the action of G proteins. Beta-3 is involved in the regulation of lipolysis and thermogenesis	p.R64W (rs4944)	Genotyping for ADRB3 revealed an Arg64 allele frequency of 5.9 versus 0.7% (HR = 11.9, $P<0.05$) compared with controls	[53]

Continued

Table 5.1 Single nucleotide polymorphisms in gallstone disease—cont'd

	Gene	Gene describe	SNP	Results	References
Hormone receptor	AR	Androgen receptor. Steroid hormone receptors are ligand-activated transcription factors that regulate eukaryotic gene expression and affect cellular proliferation and differentiation in target tissues	c.172(CAG)*n*	A significantly decreased OR for cholelithiasis risk was observed in individuals having the SL and LL genotype (OR = 0.622; 95% CI = 0.345–1.121; $P = 0.114$ and OR = 0.287; 95% CI = 0.151–0.543, $P < 0.0001$, respectively)	[54]
	ESR2	Estrogen receptor 2 (ER beta). Nuclear hormone receptor. Binds estrogens with an affinity similar to that of ESR1 and activates expression of reporter genes containing estrogen response elements (ERE) in an estrogen-dependent manner	c.1092þ3607(CA)n	A significantly decreased OR for cholelithiasis risk was observed in individuals having the SL and LL genotype for ER beta gene compared with SS genotype (OR = 0.212; 95% CI = 0.105–0.426; $P < 0.0001$ and OR = 0.042; 95% CI = 0.018–0.097, respectively)	[54]

phospholipids, or both, are decreased. Increased cholesterol saturation index (CSI) observed in patients with cholesterol gallstones may result from increased biliary cholesterol secretion. Although biliary cholesterol supersaturation is necessary, it is insufficient for gallstone formation because most individuals with supersaturated bile lack stones [68]. Cholesterol gallstone formation requires a mucin gel as nucleation matrix for cholesterol monohydrate crystals from supersaturated gallbladder bile [69]. This most often occurs in a gallbladder with impaired motility [70,71]. Cholesterol hypersaturation of bile may stimulate pathologic changes in the gallbladder epithelium thus inducing abnormal production and secretion of mucin. These factors may generate cholesterol micelles and monohydrate crystals with subsequent gallstone formation (Fig. 5.1). Mucins affect gallstone formation at the genetic level [50,51]. Defects of fasting and/or postprandial gallbladder motility associated with excess and fast crystallization of biliary cholesterol are key factors for cholesterol gallstone formation [72,73].

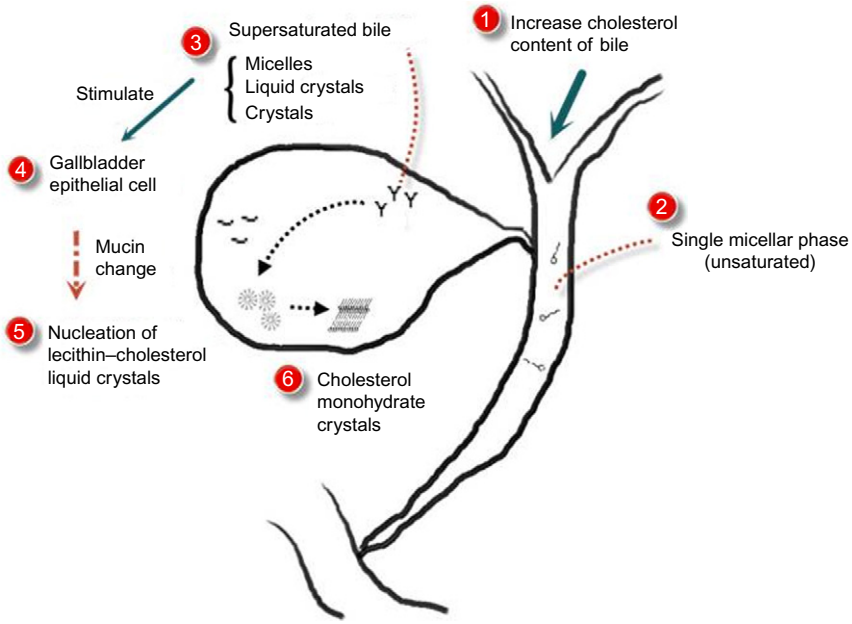

Figure 5.1 A diagram indicating the general process of cholesterol crystal formation from supersaturated bile.

3. CANDIDATE GENES INVOLVED IN REGULATION OF BILIARY CHOLESTEROL SECRETION

Cholesterol supersaturation in bile is an important pathophysiologic factor in the formation of cholesterol gallstones. Cholesterol is regulated by intestinal absorption, hepatic uptake and synthesis, and secretion into the biliary tract. In humans, most cholesterol destined for biliary secretion is derived from two sources: HDL that transports cholesterol from peripheral organs to the liver [74] and dietary cholesterol transported to the liver in chylomicron remnants rich in cholesteryl esters [75]. Hepatic cholesterol derived from LDL transport and *de novo* synthesis plays a minor role [76]. Candidate genes in biliary cholesterol regulation are summarized (Table 5.2).

3.1. Intestinal cholesterol influx and efflux

Diet and intestinal absorption contribute to subsequent cholesterol gallstone formation [19]. Absorption is a multistep process regulated by multiple genes at the enterocyte level. Animal studies indicate that delayed or impaired small intestinal transit time in the CCK-1R-null mice enhances intestinal cholesterol absorption [20]. High absorption efficiency enhances biliary secretion ultimately promoting cholesterol gallstone formation [100]. Absorption derived from dietary sources and biliary cholesterol (enterohepatic recycling) takes place in the small intestine. Net cholesterol absorption is the result of uptake and active secretion to the lumen. This process depends on active transport via an ABC transporter and passive efflux of cholesterol to intestinal epithelial cells. Enzymes and transport proteins involved in enterohepatic recycling are shown in Fig. 5.2.

Niemann–Pick C1-like protein 1 (NPC1L1) and adenosine triphosphate (ATP)-binding cassette (ABC) cotransporters (ABCG5/G8) are membrane-bound proteins expressed in the brush border of jejunal enterocytes. They actively facilitate influx and efflux of cholesterol between the lumen of the intestine and the brush border membrane of the enterocyte, respectively. Intestinal cholesterol uptake and absorption was significantly reduced (90%) in NPC1L1 null mice [77]. ABCG5/G8 is thought to export phytosterols into the bile and back to the intestinal lumen for excretion [79]. Overexpression of ABCG5/G8 genes in the intestine increases biliary secretion and small intestinal efflux in transgenic mice, whereas targeted deletion of the ABCG5 and ABCG8 genes induces a significant decrease in biliary

Table 5.2 Human candidate genes implicated in the development of cholesterol gallstone

	Gene/locus	Mouse model	Gene variants	Function	References
Cholesterol					
Intestinal cholesterol influx/efflux	NPC1L1/7p13	+		Uptake of cholesterol from the intestinal lumen to the brush border membrane of the enterocyte	[77,78]
	ABCG5/2p21, ABCG8/2p21	+/*Lith*9	ABCG8 p.D19H (rs1188753)	Promotes active efflux of cholesterol and plant sterols from the enterocyte into the intestinal lumen	[79,80]
	ABCA1/9q22–q31			Acts as a cholesterol efflux pump in the cellular lipid removal pathway	[81,82]
	ABCB1/7q21.12	+		Promotes cholesterol uptake from micelles; drug efflux pump	[83]
Plasmatic transport	ApoA1/11q23		−75G > A, RFLP (*Pst*I, *Msp*I)	An antinucleation agent in the formation of gallstones; secondary to reverse cholesterol transport	[84,85]
	ApoB/2p24	+	c.2488C > T, c.4154G > A (*Xba*I, *Eco*RI)	A ligand for the LDL receptor; secondary to hepatic VLDL synthesis↓ and intestinal cholesterol absorption↑	[39]
	ApoC1/19q13.2		RFLPs	Inhibition of CETP	[37,86]
	ApoE/19q13.2			Acts as a ligand for VLDL, mediating the plasmatic response to dietary cholesterol consumption	[28,87]
	CETP/16q21	–	RFLPs (*Eco*NI, *Taq*IA, *Taq*IB)	Hepatic cholesterol uptake from HDL catabolism	[88]

Continued

Table 5.2 Human candidate genes implicated in the development of cholesterol gallstone—cont'd

	Gene/locus	Mouse model	Gene variants	Function	References
Uptake	SR-BI/12q24.31			Promotes HDL cholesteryl ester uptake into hepatocytes	[89]
	LRPAP1/4p16.3	+		Hepatic cholesterol uptake from chylomicron remnants	[42]
Synthesis	HMG-CoA reductase/ 5q13.3–14	+		Key enzyme in the *de novo* cholesterol synthesis pathway	[90]
	ACAT1/11q22.3–q23.1 ACAT2/6q25.3–q26	+		Excess intracellular cholesterol esterification	[65,91]
	LXR/12q13.3			Regulates the metabolism of cholesterol and bile salts	[92,93]
Bile salt					
Synthesis	CYP7A1/8q11–q12	+	Promoter SNP −204A > C	Rate-limiting enzyme of bile salt synthesis	[94]
	CYP8B1/3p21.3–p22			Controls the ratio of cholic acid and chenodeoxycholic acid synthesis	[94]
	CYP27A1/2q33-qter			Initiates the alternative pathway of primary bile acid synthesis	[94]
	CYP7B1/8q21.3			Regulates the rate of cholesterol secretion into bile	[94]

	LXR/12q13.3			Upregulation of cyp7a1	[95]
	FXR/12q23.1			Downregulation of cyp7a1	[96]
Secretion	ABCB11/2q24	+/*Lith*1	Multiple	Hepatocanalicular bile salt export pump	[97]
Phospholipid					
Secretion	ABCB4/7q21.1	+	Multiple	Hepatocanalicular phosphatidylcholine (lecithin) flippase	[17,98]
Gallbladder					
Hypomotility	CCK1R/4p15.1–p15.2	+	RFLP	Gallbladder and small intestinal hypomotility	[20,99]
	ADRB3/8p12		p.R64W (rs4944)	Gallbladder hypomotility	[53]

Abbreviations: NPC1L1, Niemann–Pick C1-like 1; ABCG5/G8/A1/B1/B4/B11, ABC transporters G5/G8/A1/B1/B4/B11; ApoA1/B/C1/E, apolipoprotein A1/B/C1/E; CETP, cholesterol ester transfer protein; SR-BI, scavenger receptor class B type I; LRPAP1, low-density lipoprotein receptor; HMG-CoA reductase, 3-hydroxy-3-methylglutaryl-coenzyme A reductase; ACAT, acetyl-coenzyme A acetyltransferase; LXR, liver X receptor; CYP7A1, cholesterol 7 alpha-hydroxylase; CYP8B1, sterol 12 α-hydroxylase; CYP27A1, sterol 27-hydroxylase; CYP7B1, 25-hydroxycholesterol 7 alpha-hydroxylase; FXR, farnesoid X receptor; CCK1R, cholecystokinin 1 receptor; ADRB3, β3 adrenergic receptor; HDL, high-density lipoprotein; RFLP, restriction fragment length polymorphism; rs, restriction site; SNP, single nucleotide polymorphism; VLDL, very-low-density lipoprotein.

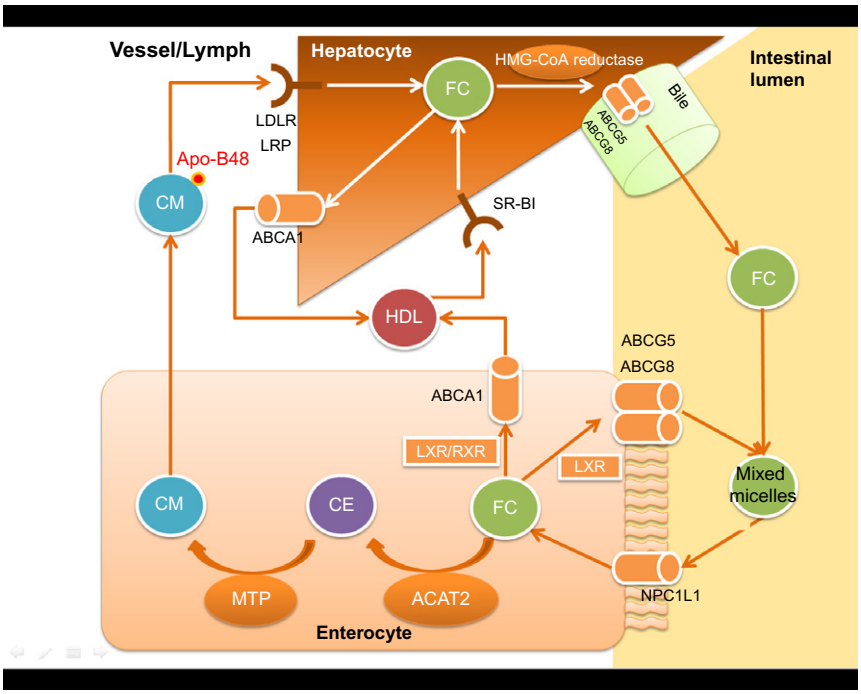

Figure 5.2 The major enzymes and transport proteins involved in the synthesis and transport of cholesterol.

cholesterol secretion in knockout mice [80]. The combined regulatory effects of NPC1L1 and ABCG5/G8 may play a critical role in modulating cholesterol movement from the intestinal lumen to enterocytes [101].

ABC subfamily A member 1 (ABCA1) functions as a cholesterol efflux pump in the cellular lipid removal pathway [81]. ABCA1 is expressed in some hepatocytes and macrophages in the lamina propria of intestinal villi. ABCA1 mRNA is not detected in the epithelial cells lining the small intestine [102]. Increased intestinal expression of ABCA1 and simultaneous inhibition of cholesterol absorption was, however, observed in mice fed a retinoid X receptor (RXR) agonist [82]. This finding suggested that these RXR heterodimers serve as key regulators of cholesterol homeostasis by governing reverse cholesterol transport from peripheral tissues, bile acid synthesis in liver, and cholesterol absorption in the intestine.

ABC subfamily B member 1 (ABCB1), also known as multidrug resistance protein 1 (MDR1) or P-glycoprotein 1 (Pgp-1), is a plasma membrane ABC transporter. In an intestinal cell line study, overexpression of ABCB1

resulted in increased cholesterol uptake from micelles [83]. In humans, ABCB1 was expressed in a number of tissues including the apical membrane of the enterocytes. It also acts as an ATP-dependent drug efflux pump for xenobiotic compounds.

3.2. Plasma transport of cholesterol

Apolipoprotein A1 (ApoA-1) is the major protein component of plasma HDL. This protein promotes cholesterol efflux from tissues to the liver for excretion. ApoA-1 knockout mice have low plasma HDL cholesterol [84] and their rate of hepatic cholesterol synthesis was 50% lower than that of wild-type mice. In contrast, ApoA-1 overexpressing mice have been reported to have a twofold increase in biliary output of bile acid and cholesterol. In humans, it has been observed that ApoA1 removes certain lipids from the bile and acts as an antinucleation agent in gallstone formation [85].

Apolipoprotein B (ApoB) is the primary apolipoprotein of chylomicrons and LDL. There are two main forms: ApoB48 and ApoB100. ApoB acts as a ligand for the LDL-R mediated by endocytosis. The ApoB gene *Xba*I polymorphism is associated with alterations in serum lipids, that is, a risk factor for cholesterol GSD. Han *et al.* studied the polymorphism of the ApoB gene *Xba*I in patients with gallstones and found that the X+/− genotype resulted in a significant increase in serum total cholesterol, LDL, and ApoB versus the X−/− genotype [39]. This finding suggested that ApoB may be a marker for increased cholesterol GSD risk.

Apolipoprotein C1 is a protein component of lipoproteins primarily expressed in liver. Its main function is the inhibition of CETP [86] via altered electrostatic HDL charge. Dixit *et al.* demonstrated that ApoC1 *Hpa*I polymorphism was significantly associated with GSD. The frequency of H2H2 was significantly higher ($P=0.017$) in patients versus controls and was associated with high risk (OR = 9.416) for GSD [37].

Apolipoprotein E, a main chylomicron component, binds the high-affinity ligand for the LDL receptor and LDL receptor-related protein on liver and peripheral cells involved in hepatic uptake of chylomicron cholesterol. ApoE knockout mice fed a high cholesterol lithogenic diet had a markedly lower frequency of gallbladder bile cholesterol crystal and gallstone formation than wild-type mice [87]. This finding likely results from decreased CSI in gallbladder bile of ApoE-deficient mice. Human defects in apolipoprotein E result in familial dysbetalipoproteinemia or HLP III. These disease states are characterized by impaired clearance of chylomicron and VLDL remnants as a consequence of increased plasma cholesterol and triglycerides.

CETP, also called plasma lipid transfer protein, is a plasma protein that facilitates transport of cholesteryl esters and triglycerides between lipoproteins. CETP collects triglycerides from VLDL or LDL and exchanges them for cholesteryl esters from HDL [103]. Juvonen *et al.* evaluated the distribution of CETP TaqIB polymorphism in patients with cholesterol gallstones versus normal subjects [88]. B1B1 subjects differed significantly ($P = 0.036$) thus suggesting that CETP gene polymorphism may be associated with cholesterol GSD.

3.3. Cholesterol uptake by hepatocytes

HDL, as the major cholesterol carrier of the reverse cholesterol transport between the periphery and liver, undergoes several modifications including the transfer of cholesteryl esters between lipoproteins by CETP and scavenger receptor class B member I (SR-BI). Hepatic uptake of esterified and free cholesterol following binding of HDL particles to the SR-BI is an important source of cholesterol for biliary secretion [104]. The SR-BI promotes the selective uptake of HDL cholesteryl ester into hepatocytes [89]. Overexpression of SR-BI following infection with recombinant adenoviruses or in transgenic mice reduced plasma HDL cholesterol and was associated with a substantial increase in the cholesterol content of gallbladder and hepatic bile, whereas bile acid or phospholipid were unchanged [104,105].

LDL receptor-related protein associated protein (LRPAP1) is a chaperone protein involved with trafficking of LDL receptor family members including LRP1 and LRP2. LRPAP1 inhibits the binding of all known ligands for these receptors and may prevent receptor aggregation and degradation in the endoplasmic reticulum thereby acting as a molecular chaperone [106]. Dixit *et al.* found that the frequency of I allele of LRPAP1 was significantly higher in the gallstone group versus controls ($P = 0.003$), suggesting that the I allele was associated with GSD [42].

3.4. Cholesterol synthesis

Liver cholesterol is derived from dietary and biliary cholesterol, *de novo* synthesis, and transport from extrahepatic tissues [107,108]. HMG-CoA reductase, an integral endoplasmic reticulum protein, is rate-limiting for *de novo* cholesterol synthesis. Gene expression and enzymatic activity are regulated by intracellular cholesterol concentration via cholesterol-derived oxysterols. When fed with a high cholesterol diet, the expression of hepatic HMG-CoA reductase was downregulated in gallstone-resistant AKR/J mice, whereas it remained unchanged in gallstone-susceptible C57L/J mice [90].

Interestingly, there is no therapeutic role for HMG-CoA reductase inhibitors in the treatment of human cholesterol GSD [109].

Excess cholesterol is esterified by acetyl-coenzyme A acetyltransferase (ACAT) [110], a group of coenzymes involved in the metabolism of fatty acids, and stored as cholesteryl esters in cytosolic droplets or secreted into the circulation as lipoprotein particles [101]. Two ACAT genes have been identified, ACAT1 and ACAT2 [111,112]. The former is found in macrophages and adrenocortical cells. The latter is restricted to the liver and intestine in mice. ACAT2 deficiency has reduced cholesterol-esterifying activity in mice fed a cholesterol-rich diet, including complete resistance to diet-induced hypercholesterolemia and cholesterol gallstone formation [65]. In the human liver, ACAT-1 is present in both fetal and adult hepatocytes, whereas ACAT-2 is present in fetal hepatocytes only [91]. As such, ACAT-1 may play the more important role in hepatic cholesterol esterification.

3.5. Biliary cholesterol and bile salt secretion

The principal driving forces for biliary lipid secretion are ABC transporters located on the canalicular membrane of the hepatocyte. As these ABC transporters are responsible for the secretion of the main lipid components of bile (cholesterol, bile acid, and phosphatidylcholine), changes in their expression and/or activity may result in altered bile composition, which may influence gallstone formation. The secretion of cholesterol into bile depends on the action of the ABCG5/G8 heterodimer [113]. ABCB11 is the dedicated common bile salt export pump (BSEP) [97,114]. ABCB4 involves transport of phospholipids from hepatocytes into bile [98]. Overexpression of the ABCG5/G8 transporter increases the cholesterol content of gallbladder bile [113]. However, gallstone formation is observed in ABCG5 or ABCG8 knockout mice challenged with a lithogenic diet [34], suggesting the existence of an ABCG5/G8-independent pathway for biliary cholesterol secretion. A network of nuclear receptors forms heterodimers with the RXR, thus regulating cholesterol and bile salt metabolism [92]. The nuclear lipid receptors include the liver X receptor (LXR) encoded by NR1H3. LXR is activated by oxysterols. The LXR/RXR heterodimer induces expression of ABCG5/G8 and consequently increases secretion of cholesterol into the bile [93].

3.6. Bile acid synthesis and transport

Hepatic cholesterol may be converted to bile acids or secreted directly into the bile via reverse cholesterol transport. In humans, the pool consists of "primary" cholic acid (CA) and chenodeoxycholic acid (CDCA) and "secondary" deoxycholic acid and lithocholic acid bile acids [115]. Primary

bile acids are synthesized from cholesterol via the classic and alternative hepatic pathways. In contrast to CA, CDCA may reduce the degree of cholesterol bile saturation thus decreasing gallstone formation. In the classic pathway, cholesterol 7 α-hydroxylase (CYP7A1) and microsomal sterol 12 α-hydroxylase (CYP8B1) are rate-limiting for CDCA and CA synthesis, respectively [92]. In addition, mitochondrial sterol 27-hydroxylase (CYP27A1) initiates the alternative pathway in both CA and CDCA synthesis [94,116]. Secondary bile acids are derived from primary bile acids by intestinal bacterial enzymes (Fig. 5.3). LXR upregulation of CYP7A1 may promote reverse cholesterol transport from the peripheral tissues to the liver for increased bile acid synthesis [95]. The nuclear hormone farnesoid X receptor (FXR), encoded by NR1H4, induces ABCB11 expression and downregulates CYP7A1 [93,96]. This effectively reduces hepatic bile salt synthesis and thus protects the liver from toxic bile acid accumulation. Multiple pathways are involved in the synthesis of bile salts. LXR and FXR play critical roles in coordinated control of bile acid, cholesterol, and triglyceride metabolism to maintain lipid homeostasis [117].

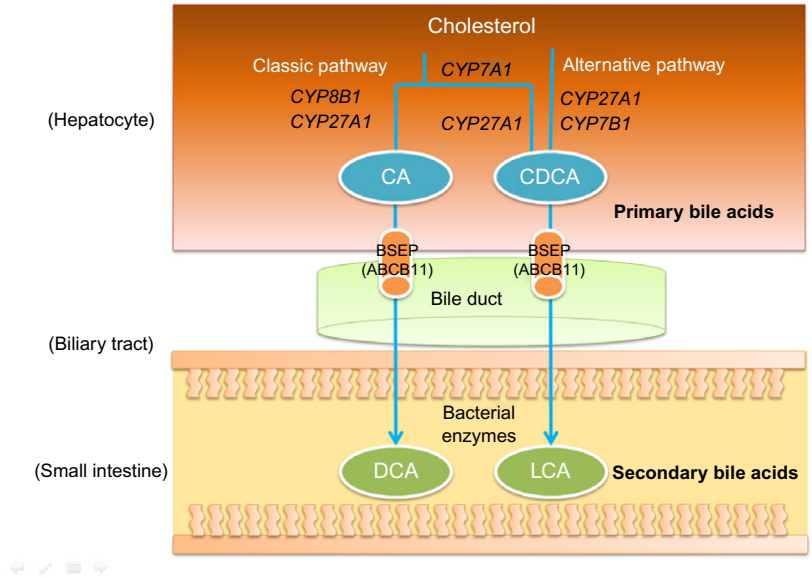

Figure 5.3 The major enzymes and transport proteins involved in the synthesis of bile acid.

The bile salt export pump (BSEP, or SPGP, or ABCB11) regulates the secretion of bile salts into bile. BSEP expression is restricted to the canalicular membrane of hepatocytes. In BSEP −/− mice, biliary bile salt output was reduced by 70% [97]. Plasma and hepatic bile acid concentrations were increased. The concentration and output of both phospholipids and cholesterol in bile were both increased.

Cholesterol, phospholipids, bile salts, and the bile pigment bilirubin are major bile components [118]. Decreased output of bile salts may induce cholesterol gallstone formation. Bile salts are efficient detergents by "enterohepatic circulation," which includes intestinal uptake by the sodium-dependent bile salt transporter ASBT (encoded by SLC10A2) and the basolateral hepatocellular uptake by the sodium/taurocholate cotransporting polypeptide NTCP (encoded by SLC10A1) [92,119].

3.7. Phospholipid transporter

Biliary lipids mainly consist of cholesterol and phospholipid synthesized in hepatocytes [92,120]. Biliary phospholipids, primarily phosphatidylcholine from *de novo* synthesis, are secreted through hepatocyte membranes into the bile canaliculi by the multiple drug resistance protein 3 (MDR3) [98]. MDR3 is encoded by ABCB4, another member of the ABC-dependent transporter family, located on hepatocyte canalicular membranes. Disruption of the ABCB4 gene in mice results in the absence of bile phospholipids [121]. However, the expression and activity of ABCB4 can be regulated by bile salts. Missense mutations in the ABCB4 gene may result in cholelithiasis characterized by intrahepatic sludge, cholesterol gallstones, mild chronic cholestasis, a high cholesterol/phospholipid ratio in bile, and recurrent symptoms after cholecystectomy [17,122]. Decreased ABCB4 expression occurs with markedly reduced bile phospholipid concentration [18]. HMG-CoA reductase activity is increased and CYP7A1 activity is reduced in gallstone patients. Mice homozygous for the MDR2 Pgp gene completely lack bile phospholipid [98]. This causes segmental biliary strictures due to periductal fibrosis, fibro-obliteration of bile ducts, and spontaneous gallstone formation [123].

4. MUCIN GENE

4.1. Mucus, mucin, and MUC genes

Mucus is composed of glycoproteins, inorganic salts, and water secreted from mucosal surfaces. It lubricates epithelial surfaces and acts as a diffusion barrier against hazardous and noxious substances [124]. Mucins are the major

component of mucus responsible for its specific gel properties to protect and lubricate the epithelium [125]. Mucin proteins are high-molecular weight epithelial glycoproteins with oligosaccharide side chains linked to tandem repeat peptides of threonine, serine, and proline [126]. In general, mucins are divided into two structurally and functionally distinct classes [127]. Membrane-bound mucins (MUC1, MUC3, MUC4, MUC12, and MUC17) and mucin-like protocadherin (MUPCDH) are primarily located at the cell surface. Gel-forming mucins (MUC2, MUC5AC, MUC5B, and MUC6) are secreted by the epithelium and provide a protective coating to the underlying mucosa. Mucin genes encode mucin monomers that are synthesized as rod-shaped apomucin cores. More than 20 mucin genes have been identified to date [128]. At least eight mucin genes (MUC1, MUC2, MUC3A, MUC3B, MUC4, MUC5AC, MUC5B, and MUC6) are involved in gallbladder function and its associated diseases (Table 5.3). Most mucin gene families reside in two distinct clusters on chromosomes 7q22 and 11p15.5 (Fig. 5.4) [129,130]. The order of the mucin genes (telomere to centromere) within 11p15.5 is MUPCDH, MUC6, MUC2, MUC5AC, and MUC5B.

4.2. Characteristics and physiologic function of gallbladder mucin

Gallbladder mucin proteins are synthesized and secreted by the surface mucous cells and submucosal glandular cells that line the epithelium. These cells can be histologically detected by periodic acid Schiff–Alcian blue

Table 5.3 Characteristics of gallbladder-associated MUC genes

Mucin gene	Alias name	Locus	DNA length (bp)	Protein length (a.a.)	Specificity
MUC1	CD227, PEM	1q21	4407	273	MBM
MUC2	–	11p15.5	29,543	5179	GFM
MUC3A	–	7q22	64,363	1486	MBM
MUC3B	–	7q22	4726	310	MBM
MUC4	–	3q29	65,209	5412	MBM
MUC5AC	MUC5	11p15.5	79,891	1962	GFM
MUC5B	MG1	11p15.5	39,112	5762	GFM
MUC6	–	11p15.5	23,883	2439	GFM
MUPCDH	CDHR5, MUCDHL	11p15.5	8503	839	MBM

Abbreviations: MBM, membrane-bound mucin; GFM, gel-forming mucin.

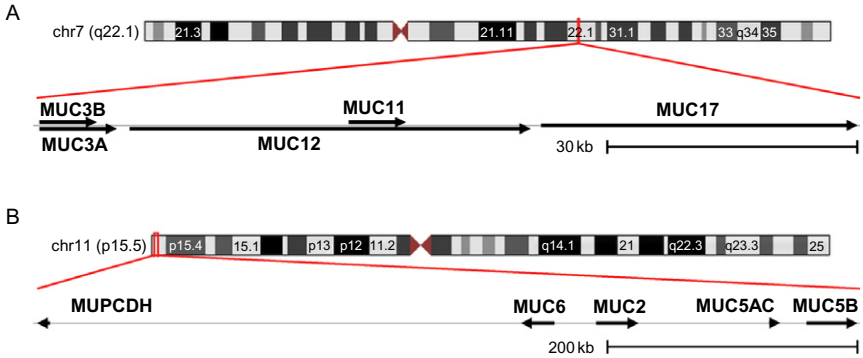

Figure 5.4 Most of the mucin gene families reside in two distinct clusters on chromosomes 7q22.1 and 11p15.5.

(PAS–AB) staining due to the presence of heavily glycosylated proteins in cytoplasmic secretary granules [131]. Of the 20 MUC genes sequenced and cloned, MUC1, MUC2, MUC3A, MUC3B, MUC4, MUC5AC, MUC5B, MUC6, and MUPCDH most likely are involved in GSD [132,133]. Gel-forming mucins (MUC2, MUC5AC, MUC5B, and MUC6) are produced by gallbladder epithelial cells. These form polymers and are responsible for the viscoelastic protective properties of mucus. Membrane-bound mucins (MUC1, MUC3A, MUC3B, MUC4, and MUPCDH) having a hydrophobic membrane-spanning domain are integral parts of the cell membrane and do not form mucous gel [134]. The gallbladder mucin protein core contains at least two distinct domains. One domain is rich in serine, threonine, and proline and contains the majority of covalently bound carbohydrates. The second domain is nonglycosylated, enriched in serine, glutamic acid, glutamine, and glycine and binds hydrophobic ligands such as bilirubin and 1-anilino-8-naphthalene sulfonate [135]. Hydrophobic domains on the mucin protein core may contribute to the pathogenesis of cholesterol cholelithiasis.

Different mucin genes are expressed in the gallbladder epithelium. The primary mucins are MUC3, MUC5B, and MUC6 [136]. MUC5B is predominant [137], even in gallbladder with stones [132]. Gallbladder mucin is composed of multiple layers [138]. MUC1 and MUC3 are membrane-bound protecting the apical surface. MUC5AC and MUC5B are secreted by surface mucous cells and constitute the next layer. MUC5B and MUC6 are secreted from deeper mucous cells. *In situ* hybridization was used to study mucin mRNA expression in 13 human embryos, fetuses, and adult gallbladders [133]. This

study found that MUC1, MUC2, MUC3, MUC5AC, MUC5B, and MUC6 were expressed in adult and fetal gallbladder, but MUC4 was not. MUC5AC and MUC5B were found in 95.4% and 100% of 16 gallbladder bile samples, respectively [134]. These mucins in the gallbladder epithelium are necessary to protect from the detergent effect of increased bile salts [134,138].

4.3. Role of gallbladder mucin in pathophysiology of gallstones

Early studies identified the importance of mucin in cholesterol gallstone pathogenesis [139,140]. Womack *et al.* [141] noted that gallbladder mucus hypersecretion preceded gallstone development in hamsters fed a lithogenic diet. This study found that cholesterol crystals first formed in aggregates of mucus rather than in the liquid phase of gallbladder bile. Englert *et al.* [142] noted increased mucus secretion during diet-induced pigment gallstones in canines. Similar animal studies on cholesterol-fed rabbits [143] and prairie dogs [144] also found increased mucin before gallstone formation. Sheen *et al.* evaluated the PAS–AB index on the lining epithelia of gallbladders [131]. This study showed that brown stones were significantly more numerous than cholesterol stones and gallbladders without stones. These findings suggested an alteration in gallbladder mucous that induced stone development. A macromolecular complex of mucin and bilirubin, within the central nidus and throughout the stone matrix, was identified as a major structural component [145]. As such, gallbladder mucin may promote cholesterol gallstone formation by accelerating cholesterol monohydrate crystal nucleation in supersaturated bile [135,146].

Mucin gene expression may be upregulated by a variety of factors including inflammatory cytokines, bacterial products, growth factors, environmental chemicals or pollutants, and the immune system [147,148]. The quantity, size, and chemical nature of mucins, especially the gel formers, can be altered, thus impacting gel properties [149]. For example, hypersecretion of bile cholesterol and chronic inflammation of the gallbladder wall have been implicated in cholesterol GSD [150]. An inflammation-dependent epidermal growth factor receptor (EGFR) cascade caused overproduction of gel-forming MUC5AC [151]. MUC5AC overexpression may also be induced by interleukin-1 beta (IL-1β) and tumor necrosis factor-alpha (TNFα) [152]. This mechanism involved ERK/p38 mitogen-activated protein kinases-MSK1–CREB activation and was mediated by cyclooxygenase-2, prostaglandin E2 receptors, and cAMP protein kinase [153]. Cholesterol monohydrate crystals induced proinflammatory cytokine expression in a T-cell-dependent fashion [154].

Acquired immunity and inflammation via MUC3 and MUC4 led to increased stone susceptibility. Gallbladder epithelium inflammation was associated with increased MUC5AC in both the superficial epithelium and deep folds [134]. Inflammatory mediators have been identified including TNFα-induced mucin hypersecretion and MUC2 gene expression [155]. Protein kinase C (PKC) mediated MUC2 overexpression [156]. IL-1β mediated MUC2 gene expression via activation of the PKC–MEK/ERK pathway in NCI-H292 cells [157]. Leukotriene D4 was found to upregulate MUC2/5AC gene expression [158]. Inflammation may also affect membrane-bound mucins. Although MUC1 mRNA was present in all human gallbladders, apoMUC1 was not immunologically found but tended to occur in specimens with increased degree of inflammation [138]. These studies all support the role of inflammation in altering mucin secretion.

A component of lithogenic bile appears to stimulate gallbladder mucin hypersecretion. Gel-forming mucins form polymers to protect the epithelium from the detergent effect of bile. However, the hydrophobic-binding sites in the polypeptide core may also provide a favorable environment for nucleation of cholesterol monohydrate crystals from supersaturated bile [159,160]. Aberrant and/or enhanced expression of mucin polypeptides may alter biliary lipid binding thus forming a mucus gel that promotes cholesterol crystallization [151]. As such, this mechanism may be very important to the lithogenic process [161]. The C-termini of MUC5AC, MUC5B, and MUC6 are cysteine-rich. These domains enable formation of end-to-end polymers responsible for gel viscosity [138]. As MUC2 and MUC5AC increase, biliary tract mucus becomes more viscous further promoting nuclear formation and stone enlargement [156]. These observations suggest that mucins display different biologic functions in the gallbladder and may have different lithogenic effects on the formation of gallstones.

4.4. The alternation of mucin expression in GSD

Alterations of mucin glycoproteins have been described in different pathologic conditions [132,162,163]. Early studies demonstrated that gallbladder bile from patients with gallstones contained more glycoproteins and was more viscous due to its greater concentration of mucus [164,165]. Gallbladder mucus hypersecretion and increased mucin concentration were also demonstrated in experimental models of stone formation [143,144,160,166]. LaMont *et al.* [167] demonstrated that mucin glycoproteins could be purified from black pigment stones.

Altered mucin gene expression is found in gallbladders with stones. Although MUC2 and MUC4 are not found normally, they are present in gallbladders with stones [168]. The presence of focal MUC2-reactive goblet cells may reflect intestinal metaplasia. Focal intestinal metaplasia may involve any part of the gallbladder in association with cholelithiasis, congenital abnormalities, and neoplasia [169–171]. Increased MUC1, MUC3, MUC5B, and MUC6 [136], and MUC5AC [151] expression has been reported in human gallbladders with GSD. In gallbladder tissue sections from subjects with gallstones, MUC2 and MUC5AC were overexpressed and potentially related to mucus hypersecretion [161]. MUC5AC and MUC6 were significantly increased in the superficial epithelium and deep folds, respectively [138]. MUC5AC was overexpressed in GSD, despite decreased EGFR mRNA expression. MUC5AC may be related to mucus hypersecretion [168].

Altered mucin gene expression also occurred in gallbladders with black pigment stones. MUC2 and MUC4 and increased MUC1, MUC3, MUC5B, and MUC6 expression were found in black [132] as well as cholesterol and calcium bilirubinate stones [136]. Increased MUC2, MUC3, MUC5B, and MUC6 gene expression in black stones may reflect abundant mucin gallbladder content. Increased MUC2 and MUC4 expression in black stone-containing gallbladder epithelium indicated that intestinal metaplasia and altered mucin genes occur in disease states [163]. Gallbladders with brown stones contain more mucin than those with cholesterol stones or without stones [131]. However, another study demonstrated no difference in mucin expression in cholesterol and pigmented stones [172]. Differential expression of mucins and their associated gallbladder diseases are summarized (Table 5.4).

The role of mucins in gallstone formation has recently been questioned [173]. Using retrograde cholangiography, this study demonstrated no change in mucin secretion and expression in GSD and was unable to support the role of mucin in the etiology of biliary sludge or stone formation. The authors do point out, however, a weakness of this study since it did not include normal subjects.

4.5. Pharmacologic inhibition of mucus secretion

Aspirin given to cholesterol-fed prairie dogs inhibited mucin synthesis, hypersecretion, and gel accumulation, and prevented cholesterol crystal development and gallstone formation without influencing the cholesterol content of supersaturated bile [174,175]. Human studies have demonstrated

Table 5.4 The expression of mucins in the gallbladder and associated diseases

Subjects	Findings	References
Native gallbladder wall	MUC1, MUC2, MUC3, MUC5AC, MUC5B, and MUC6 genes were present in adults and fetal gallbladder, whereas MUC4 was not	[133]
	MUC3, MUC5B, and MUC6 mainly	[132,137]
	MUC3, MUC5AC, and MUC5B on surface epithelium, MUC5B and MUC6 on deeper mucosal folds	[138]
Mucin in bile	MUC5AC (95.4%) and MUC5B (100%), no expression of MUC2 and MUC6	[134]
	High levels of MUC3, MUC5AC, and MUC5B	[173]
Gallstone disease		
1. Cholesterol stone	Increased expression of MUC5AC and MUC2	[161]
	Gallbladder labeling MUC2 14%, MUC4 50%, and MUC1 79%	[132]
2. Brown stone	Gallbladder labeling MUC2 25%, MUC4 58%, and MUC1 83%	[132]
3. Black stones	Presence of MUC2 and MUC4 and increased expression of MUC1, MUC3, MUC5B, and MUC6	[132]
Acute cholecystitis	MUC3, MUC5AC, MUC5B, and MUC6 decreased expression; but MUC2 increased expression	[138]
	Total mucin secretion in the gallbladder bile was not associated with a higher degree of inflammation	[134]
Hepatolithiasis	MUC5AC and MUC2 increased expression	[161]

that low-dose aspirin significantly inhibits mucus synthesis in the gallbladder [176] and may reduce gallstone recurrence [177]. Possible mechanisms for inhibition of mucin biosynthesis by aspirin include reduced prostaglandins or depletion of cellular energy [176,178].

A double-blind, placebo-controlled study showed that ursodeoxycholic acid reduced mucin concentration and cholesterol crystal formation in the gallbladder bile of patients with cholesterol gallstones [179]. This effect

might be mediated by decreased of biliary lipid peroxidation. The authors therefore suggested that inhibition of mucin release may prevent cholesterol stone formation during high risk periods or after dissolution therapy with bile salts.

4.6. Mucin genetic predisposition to GSD

Molecular studies have shown that mucin production, secretion, and accumulation are determined by multiple mucin genes [180–182]. MUC1 transgenic mice have been shown to increase risk for GSD by promoting gallbladder cholesterol absorption and impairing gallbladder motility [183]. MUC1 knockout mice had significantly decreased MUC5ac and MUC5b mRNA expression [20]. Interestingly, MUC3 and MUC4 were significantly increased. A similar study also showed a positive correlation between MUC1 expression in bile and the expression of all the other mucin genes [173] indicating a gene–gene interaction that might affect mucin gel accumulation and cholesterol gallstone formation.

Recent SNP studies demonstrated that the MUC1 (rs4072037) and MUC2 (rs7396030) were significantly associated with gallstone formation in males, but not in females [50]. Gender-specific analysis showed significance with male subjects only ($P=0.005$). The additive interaction model for males yielded an age- and BMI-adjusted OR of 4.68 ($P=0.0008$) for these SNP. The addition of one gene resulted in an approximately 4.7-fold increased risk for GSD. When borderline risk SNP rs1806440 was included, the age- and BMI-adjusted OR was 9.82 ($P=0.0001$). Another study for the MUPCDH genetic polymorphism found that SNP rs3758650 was associated with GSD at an OR of 1.59 ($P=0.013$) for the AG genotype and 5.82 ($P=0.007$) for the AA genotype versus the GG genotype [51].

5. GALLBLADDER MOTILITY

Gallbladder hypomotility is a risk factor for gallstone formation. Motility is regulated by intrinsic cholinergic neurons [184] and direct actions of neurotransmitters to myocyte surface receptors [185]. The cholinergic ganglionic plexus is intrinsic to the gallbladder wall, lying between the serosa and muscle layers. Cholecystokinin (CCK) is secreted from specialized cells in the intestine and binds its receptor (CCK-1R) on gallbladder smooth muscle cells [186]. It activates the intrinsic cholinergic postganglionic nerves [99] and directly binds neurotransmitters to CCK-1 receptors [187]. Molecular variation in the CCK-1R gene is associated with gallstone risk, and

aberrant splicing of CCK-1R, which is predicted to result in a nonfunctional receptor, is found in a few obese patients with gallstones [188,189]. Wang *et al.* [20] found a significantly higher prevalence of cholesterol gallstones in the CCK-1R-null mice.

The beta3-adrenergic receptor (ADRB3), a transmembrane receptor highly expressed in adipose and gallbladder tissue, may be involved in gallbladder contraction. Klass *et al.* [53] performed a case–control study and found that the ADRB3 Trp64Arg polymorphism is associated with GSD. In addition, muscarinic receptor knockout mice provided direct evidence that M(3) receptors are the predominant muscarinic receptors mediating gallbladder contractility and that M(4) receptors appear necessary for optimal potency of carbamylcholine in gallbladder contraction [190].

6. CONCLUSION

The etiology and pathogenesis of cholesterol gallstone formation are multifactorial, involving complex genetic and environmental factors. Genetic predisposition plays an important role in gallstone formation. Advances in genetic analysis and availability of the human genome have made it possible to identify and isolate factors involved with GSD. GWA studies have provided the most effective approach to identify genetic variants that contribute to GSD. Individual genomic risk profiles (entire set of common *LITH* genes) will provide new insight and likely elucidate novel strategies for treatment and management of GSD.

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